Zhenglin Geng

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Education

- **School of Software, Tsinghua University, Beijing (Date Degree Expected: 07/2016)** Aug 2012 - Present

B.S. in Computer Science, Major GPA: **93/100**(Ranking: **3/57**), Second Major: Economics.

Core Courses: Computer System Software (1) (**97, top 1**), Computer System Software (2) (**99, top 4**), Data Structure and Algorithm (2) (**95, top 1**), Software Theory (**99, top 2**), Numerical Analysis (**100, top 1**), Probability and Statistics (**100, top 1**), Physics for Scientists and Engineers (**100, top 1**), Web Design and Produce (**100, top 1**), Web Front-end Technology (**100, top 1**).

Awards: 2015 Meritorious Winner in Mathematical Contest in Modeling (MCM); 2015 National Endeavor Fellowship; 2015 Lixin Tang Fellowship; 2014 National Fellowship; 2013 National Endeavor Fellowship.

Research Experience

- **Composition-aware Video Cropping (Internship, MSRA)** Oct 2015 - present

Advisor: **Steve Lin**, Senior Researcher at Microsoft Research Asia

- Developing a program to generate videos with both professional composition and visually pleasing camera motion by panning, scaling and rotating a cropping window over casually taken videos.
- Proposed a method based on video segmentation to learn from professionally cropped videos made by artists.

- **Real-Time Rigid Body Simulation with Game Engine Integration (UGVR, Stanford)** Jun 2015 - Aug 2015

Advisor: **Ron Fedkiw**, Professor at Department of Computer Science

- Developed the interfaces for the communication between the physics simulation library PhysBAM (over 3K files, 3.5m lines) and the game engine Unity, thus significantly improved the rendering performance for the simulation and greatly reduced the workload in developing graphics program.
- Implemented special effects including explosions, cloth, fluid trees and articulated bridges in a real time rigid body simulation game.

- **IFace——a one-click tool for face replacement** Jan 2014-May 2014

Advisor: **Bin Wang**, Associate Professor at School of Software, Tsinghua University

- Contributed to the development of an Android application, *iFace*, which provides a natural replacement for the faces in a picture with faces taken from other pictures.
- Our app won the **Second Prize of Challenge Cup** at Tsinghua University in 2014.

- **Content Based Image Retrieval** May 2014 – May 2014

Advisor: **Xiaoming Jin**, Associate Professor at School of Software, Tsinghua University

- Realized the R-tree structure with advanced split-node method and different relevance ranking methods, and evaluated the performance of them during the image searching process.
- Compared different machine learning methods, like KNN, K-means, Naïve Bayesian, K-D tree, neural networks (BP and Radius Basis Function) in image retrieval process after the feature extraction procedure.

Miscellanea

- **Programming skills**
  - Rich project experiences in C/C++, C#, Java, Python, JavaScript and Matlab; Adept at Haskell and MASM Assembly.
  - Familiar with Visual C++, gcc, gdb and CMake; Familiar with git and svn; Familiar with Ubuntu.
  - Familiar with OpenCV; Familiar with OpenGL/GLSL and DirectX 11/HLSL; Familiar with Blender and Unity;
  - Familiar with Qt, MFC; Familiar with web development (Django, Nodejs, ASP.NET); Familiar with Android development.
- **English skills**
  - **TOEFL iBT** : 110 (Reading: 30, Listening: 30, Speaking: 23, Writing: 27).
  - **GRE**: Verbal: 163, Quantitative: 170, Writing: 4.0.
- **Welcome to my home page** http://gengzhenglin.github.io **for more projects with videos and demos!**